



AWS Service Catalog

Lecture Slides

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AWS Service Catalog



Create and **govern self-service access** to approved IT resources.



Introduction to AWS Service Catalog

AWS Service Catalog enables organizations to **create and manage catalogs of products** that are **approved for use** on AWS to achieve consistent governance and meet compliance requirements.

The AWS Service Catalog is an alternative to granting direct access to AWS resources via the AWS Console.

- **Standardization**
- **Self-service discovery and launch**
- **Fine-grain access control**
- **Extensibility and version control**



Anatomy of AWS Service Catalog

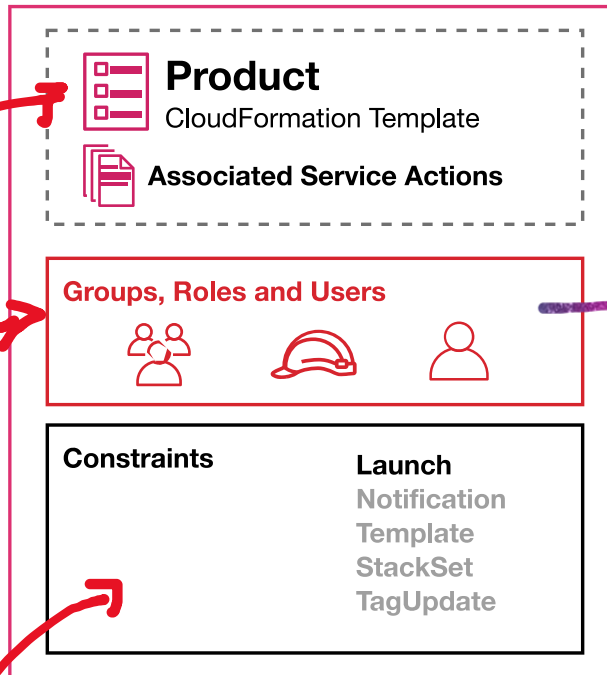
Administrative User
Manages the catalog

Portfolio is a container for

- Products
- Permissions
- Constraints

Product is a
CloudFormation
Template

Portfolio



Permissions of who
can view and launch
products

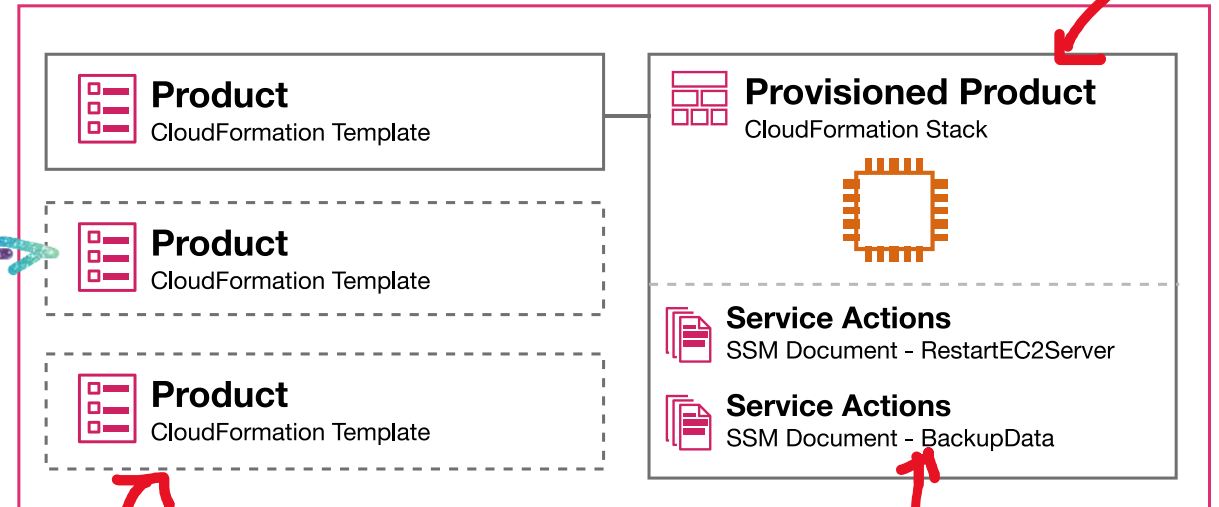
Constraints apply rules on provisioned
products. There are 5 different rules.

End User
Uses the catalog

The **Catalog** is a user-friendly console to view and launch products

Provisioned products are “**launched**” products
They are just CloudFormation Stacks

Catalog



Products are visible in the
catalog to launch

Service Actions are SSM Documents
to perform tasks on the stack



AWS Service Catalog – Users

There are two types of Service Catalog users:

1. **Catalog Administrator:** They managed the catalog
2. **End Users:** They use the catalog

Catalog Administrators

Manage a catalog of products, organizing them into portfolios and granting access to the end users.

An administrator's technical responsibilities include:

- Preparing CloudFormation templates
- Configuring constraints
- Managing IAM roles assigned to products

End Users

Use the AWS Management Console to launch products that admins have granted access to end user.



AWS Service Catalog – Administrator Product

A **product** is a **CloudFormation template** that defines the resources that will be launched

```
Resources:
  WebServer:
    Type: 'AWS::EC2::Instance'
    Properties:
      ImageId:      ami-02354e95b39ca8dec
      InstanceType: t2.micro
```

Products (1)

Search products

Product name

Created

Vendor

Description

Current vs. budget

Forecast



MinecraftServer

Thu, Aug 6, 2020, 12:58:54 PM EDT

You can create or associate an **AWS Budget** with a product

Once a product is created, it cannot be edited, only **deleted**.

Also, a product must be removed from the portfolio and not provisioned by a user in order to delete it.

Actions

Upload new product

Add product to portfolio

Copy product

Delete product

Create budget

Associate budget

Disassociate budget

In order for products to be made visible to users, it needs to be added to a portfolio.



AWS Service Catalog – Administrator Portfolio

A Portfolio is a **collection of products**.

Constraints can restrict how products are used.

To determine who can see and launch products by associating either **Groups, roles, or users**.

The screenshot shows the AWS Service Catalog Administrator Portfolio interface. At the top, there are several tabs: **Products (1)**, **Constraints (0)**, **Groups, roles, and users (1)**, **Share (0)**, **Tags (0)**, and **TagOptions (0)**. The **Products (1)** tab is selected and highlighted with a red arrow. Below the tabs, the **Products (1)** section is visible. It includes a search bar with the placeholder text "Search products", a refresh button, and three action buttons: "Upload new product", "Add product to portfolio", and "Remove". Below the search bar is a table with the following columns: Name, Id, Created time, Vendor, Provided by, and Description. The table contains one product entry: "MinecraftServer" with Id "prod-ya5zpqluuro36", Created time "Thu, Aug 6, 2020, 1:59:09 PM EDT", Vendor "ExamPro", Provided by "ExamPro", and Description "ExamPro".

	Name ▼	Id ▼	Created time ▼	Vendor ▼	Provided by ▼	Description ▼
☐	MinecraftServer	prod-ya5zpqluuro36	Thu, Aug 6, 2020, 1:59:09 PM EDT	ExamPro	ExamPro	ExamPro



AWS Service Catalog – Administrator Portfolio

To grant access so end-users can see products in the catalog, you need to associate with a portfolio **Groups**, **Users**, or **Roles**. All Products in the portfolio will be shared with the added identities. You cannot limit some products to some users in a portfolio

Groups	Roles	Users
Groups (7)		
Search groups		
☐	Group name	
☐	Admin	

Groups	Roles	Users
Roles (1/152)		
Search roles		
☐	Role name	
☒	ABRole	

Grant users access to the portfolio and allow		
Groups	Roles	Users
Users (5)		
Search users		
☐	User name ▼	
☐	bayko	
☐	brown	



AWS Service Catalog – Administrator Constraints

You can create **Constraints** for specific products in your portfolio

Product

Select the product from the portfolio to apply the constraints to.

MinecraftServer ▼

Constraint type

Choose the type of constraint to apply to the product you've selected.



Launch

Allows you to assign an IAM role to the product that is used to provision the AWS resources.



Notification

Allows you to stream product notifications to an Amazon SNS topic.



Template

Allows you to limit the options that are available to end users when they launch the product.



StackSet

Allows you to configure product deployment across accounts and regions using AWS CloudFormation StackSets.



Tag Update

Allows you to update tags after the product has been provisioned.

Launch

Use a specified IAM role instead the end-user credentials

This way, you don't have to grant the end-user permissions to services directly, which will be less permissive and more secure.

Notifications

Send product notifications to a stack

Template

Limit the options provided to the end user when launching a product. Set restrictions on the underlying CloudFormation parameter inputs. eg. Only allow t2.micro

StackSet

Allows you to configure product deployment across accounts and regions using AWS CloudFormation StackSets.

TagUpdate

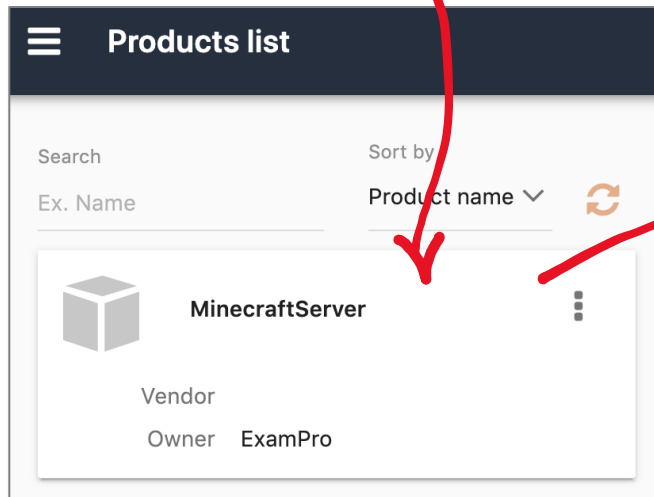
choose to allow or disallow your end users to update tags on resources associated with a provisioned product



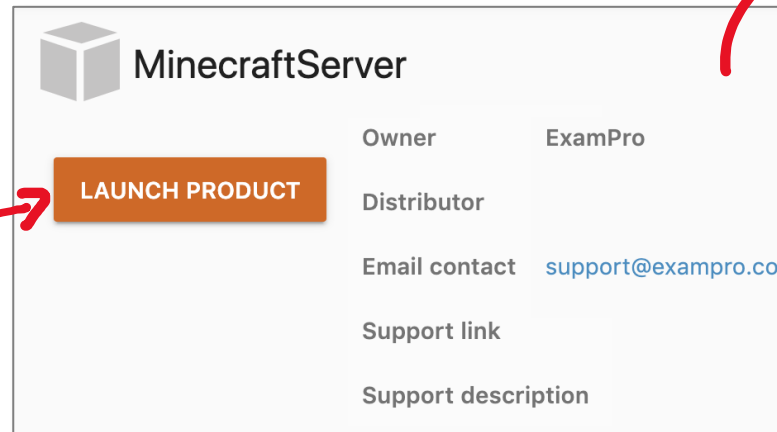
AWS Service Catalog – End User Products

The end user experiences a user-friendly interface in the AWS Console when opening up AWS Service Catalog. They can see two things, **Products**, and **Provisioned Products**

1. They can see a **catalog** of products



2. **Review Details** and **Launch** the Product



3. **Name** and **choose** the product version.

Version*		
By name	Provided by	Created time
1.0.0	ExamPro	Aug 6th 2020 13:59:09 ...

*Required CANCEL NEXT

4. Fill in the **parameters** (specified in the CloudFormation template) and finish launching.

Specify values or use the default values for the parameters.

Parameter	Value
AdminUsername	andrewbrown
AdminPassword	Admin's Password
ServerUserLimit	10
MinecraftVersion	1.16

CANCEL PREVIOUS NEXT



AWS Service Catalog – End User Products

Before launching your Product, you can choose to create a **Provisioned Product Plan**.
A plan includes a list of resources that will be created or modified before the product is launched.

▼ Resource changes

CREATE PLAN

To view resource changes, you must create a provisioned product plan. Once you create a plan, you won't be able to execute any actions until the plan is available.

▼ Resource changes

Action	Logical resource id	Physical resource id	Resource type	Replacement
Add	WebServer		AWS::EC2::Instance	

CANCEL PREVIOUS LAUNCH

This is essentially a CloudFormation ChangeSet



AWS Service Catalog – End User Provisioned Products

Provisioned Products

A product has been launched, and we can monitor it.

Provisioned products list

Search

Ex. Name OR status:available

Filter by

User

Sort by

Relevance

Count

100

Showing 1-1

< 1 >

MyMinecraftServer

Type CFN_STACK

ID pp-ymky4pt4lbhce

Created 2020-08-06 14:18:48 UTC-0400

Status AVAILABLE

Status message

MyMinecraftServer

ACTIONS

Internal App

Status Available

Product MinecraftServer

Version 1.0.0

Provided by ExamPro

Events (2)

Aug 6th 2020

Status

Type

Event message

14:20:37 UTC-0400

Succeeded

PROVISION_PRODUCT

Record ID:rec-cpufys4kc4k7q

Provisioned product ID:pp-ymky4pt4lbhce

Outputs:

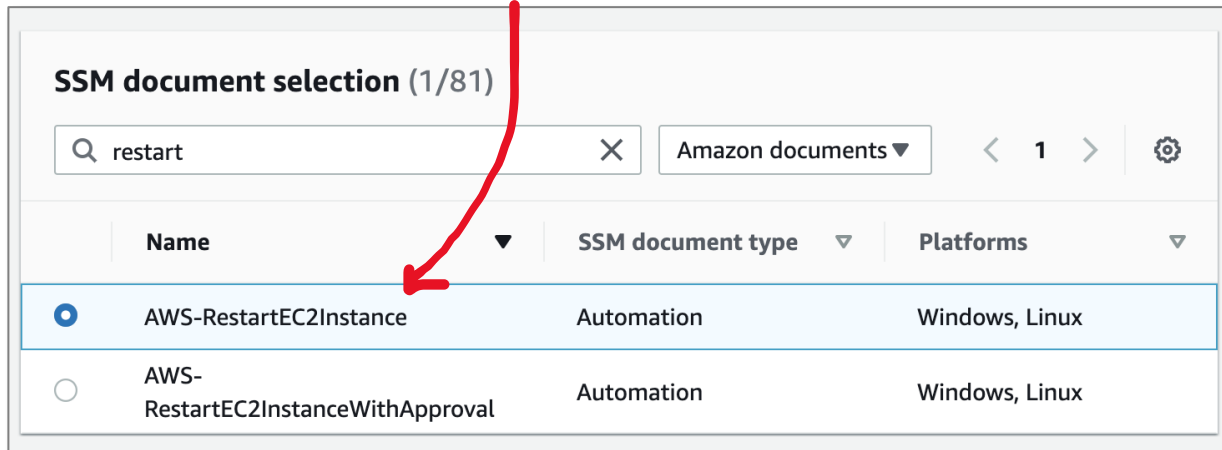
Key	Value	Description
CloudformationStackARN	arn:aws:cloudformation:us-east-1:123456789012:stack/S-C-655604346524-pp-ymky4pt4lbhce/46dcf1b0-d811-11ea-922f-0a0207fddeaf	The ARN of the launched Cl...



AWS Service Catalog – Service Actions

Service Actions are **SSM Documents associated with a Product** to allow the end user to perform maintenance.

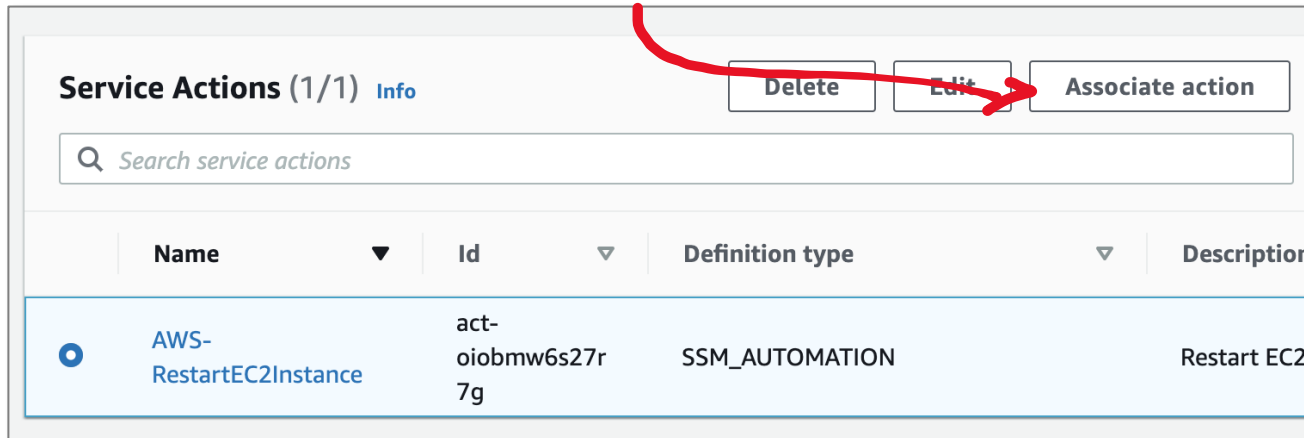
1. Administrator user **chooses** an SSM Document and create a service action



The screenshot shows the 'SSM document selection (1/81)' interface. A search bar contains the text 'restart'. Below the search bar is a table with columns: Name, SSM document type, and Platforms. Two documents are listed: 'AWS-RestartEC2Instance' and 'AWS-RestartEC2InstanceWithApproval'. A red arrow points from the 'Name' column header to the first document row.

Name	SSM document type	Platforms
☒ AWS-RestartEC2Instance	Automation	Windows, Linux
☐ AWS-RestartEC2InstanceWithApproval	Automation	Windows, Linux

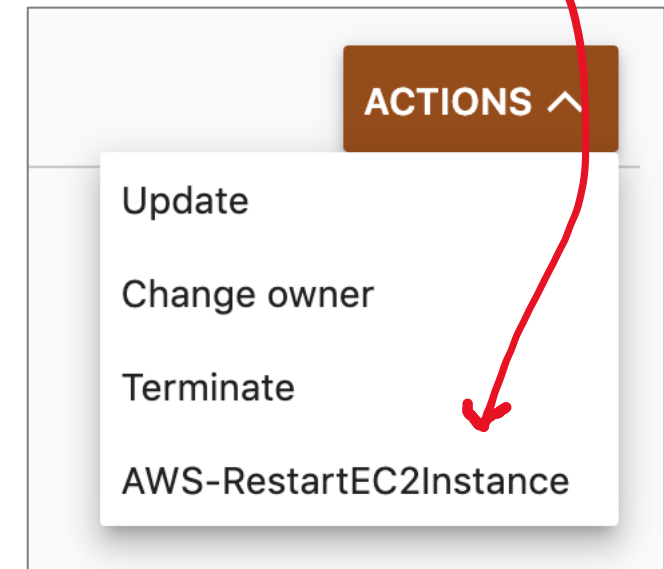
2. Administrator user **associates** the action to a product



The screenshot shows the 'Service Actions (1/1)' interface. At the top, there are buttons for 'Delete', 'Edit', and 'Associate action'. A red arrow points from the 'Associate action' button to the 'Id' column of the table below. The table has columns: Name, Id, Definition type, and Description. One service action is listed: 'AWS-RestartEC2Instance' with Id 'act-oiobmw6s27r7g' and Definition type 'SSM_AUTOMATION'.

Name	Id	Definition type	Description
☒ AWS-RestartEC2Instance	act-oiobmw6s27r7g	SSM_AUTOMATION	Restart EC2

3. The service action will appear on the Provisioned Product and the end-user can **run** the action.



The screenshot shows a dropdown menu titled 'ACTIONS ^'. The menu contains the following options: 'Update', 'Change owner', 'Terminate', and 'AWS-RestartEC2Instance'. A red arrow points from the 'AWS-RestartEC2Instance' option to the right.

- Update
- Change owner
- Terminate
- AWS-RestartEC2Instance



Service Catalog *CheatSheet*

- **AWS Service Catalog** enables organizations to **create and manage catalogs of products** that are **approved for use** on AWS to achieve consistent governance and meet compliance requirements.
- There are two types of Service Catalog users:
 1. **Catalog Administrator**: They managed the catalog
 2. **End Users**: They use the catalog
- A **product** is a **CloudFormation template** that defines the resources that will be launched
 - You can create or associate an **AWS Budget** with a product
 - Once a product is created, it cannot be edited, only **deleted**.
 - Also, a product must be removed from the portfolio and not provisioned by a user in order to delete.
 - In order for products to be made visible to users, it needs to be added to a portfolio.
- A **Portfolio** is a **collection of products** with the following options:
 - **Constraints** can restrict how products are used.
 - **Permissions** determine who can see and launch products by associating either **Groups, roles, or users**
 - You cannot restrict certain products to certain users in a portfolio, you have to just make additional portfolios
- When a Product is launched, it's called a **Provisioned Product**
- You can attach **Service Actions** (Which are just SSM Documents) to a product so end-users can perform governed actions on the product.